



Early functional recovery following accelerated rehabilitation after arthroscopic anterior talofibular ligament repair

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ABSTRACT

Background: Postoperative rehabilitation following anterior talofibular ligament (ATFL) repair commonly incorporates an initial period of protection with progressive weight-bearing and range of motion. Accelerated rehabilitation may facilitate earlier functional recovery, but clinical evidence remains limited.

Methods: A retrospective cohort study was performed on consecutive patients who underwent arthroscopic ATFL repair for chronic lateral ankle instability at a tertiary institution between January 2024 and June 2025. All patients followed a standardised accelerated rehabilitation protocol that included immediate weight-bearing and supervised range-of-motion exercises after surgery. Outcomes assessed included Foot and Ankle Ability Measure (FAAM) scores, Visual Analogue Scale (VAS), time to return to jogging, return to pre-injury level of sport and documented postoperative complications.

Results: Twenty patients were included, with a mean follow-up of 11.1 ± 3.5 months. Median FAAM Activities of Daily Living scores improved from 76.2 (Q1–Q3, 67.0–83.0) preoperatively to 99.2 (95.9–100.0) at 6 months, while median FAAM Sports scores improved from 43.5 (32.2–63.4) to 94.7 (82.8–100.0) (both $p < 0.001$). Median VAS scores decreased from 5.0 (4.0–6.0) preoperatively to 0.0 (0.0–1.0) at 6 months ($p < 0.001$). The median time to return to jogging was 42 days. Seventeen patients participated in sport preoperatively, all of whom returned to their pre-injury level of sport (17/17, 100%) at a median of 8 weeks (Q1–Q3, 6–11 weeks). No recurrent instability, repair failure, wound complication, or reoperation was observed.

Conclusion: Accelerated rehabilitation following isolated arthroscopic ATFL repair was associated with significant early functional improvement and return to pre-injury sporting level in all preoperatively active patients, with no repair failures or recurrent instability observed during short-term follow-up.

Level of Evidence: IV

1. Introduction

Chronic lateral ankle instability, a common sequela of lateral ankle sprains, may occur in approximately 20–40% of patients following an initial injury. While many patients improve with conservative management, those with persistent symptomatic instability often require surgical stabilisation [1–3]. Surgical repair of the anterior talofibular ligament (ATFL) has demonstrated reliable restoration of ankle stability and functional outcomes, with high rates of return to activity [4,5].

Postoperative rehabilitation protocols following ATFL repair vary considerably. Many protocols incorporate an initial period of protection using bracing or external support, restriction of potentially provocative

ankle movements, and progressive advancement of weight bearing and strengthening to protect the repair during early healing [6–8]. However, the optimal timing of weight-bearing, range of motion, strengthening, and progression to higher-level functional activity remains uncertain.

Arthroscopic techniques for chronic ankle instability have demonstrated favourable clinical outcomes in several settings. Vega et al. reported outcomes following combined arthroscopic lateral and medial ligament repair for rotational ankle instability, while Guelfi et al. compared arthroscopic-assisted and all-arthroscopic ankle stabilisation and subsequently reported mid- to long-term outcomes following arthroscopic all-inside repair. These studies principally evaluated surgical techniques and clinical outcomes rather than a standardised

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accelerated rehabilitation pathway [9–11].

Clinical evidence specifically evaluating accelerated rehabilitation following arthroscopic lateral ankle ligament repair remains limited. Martin et al. reported favourable early outcomes following an accelerated rehabilitation protocol using a suture-tape-augmented modified Broström procedure [12]. Teramoto et al. subsequently reported early return to sport following accelerated rehabilitation after arthroscopic lateral ankle ligament repair [13]. However, there remains limited evidence describing serial early patient-reported functional recovery and return-to-activity outcomes following a standardised, criterion-based accelerated rehabilitation protocol after isolated arthroscopic ATFL repair without suture tape augmentation.

The purpose of this study was to evaluate the early functional outcomes and safety profile of a standardised, criterion-based accelerated rehabilitation protocol following isolated arthroscopic ATFL repair without suture tape augmentation. We hypothesised that patients undergoing accelerated rehabilitation would demonstrate significant improvements in pain and function, early return to jogging, return to pre-injury level of sport, and a low incidence of short-term complications.

2. Materials and Methods

This retrospective cohort study was reviewed by the institutional review board, which waived the requirement for formal ethical approval and informed consent.

2.1. Patient Selection

Patients who underwent arthroscopic ATFL repair for chronic lateral ankle instability between January 2024 and June 2025 were retrospectively identified. All eligible patients during this period were included consecutively.

Chronic lateral ankle instability was diagnosed based on a history of recurrent ankle sprains and/or symptomatic episodes of giving way, persistence of symptoms for more than 3 months despite structured nonoperative rehabilitation, and clinical evidence of lateral ankle instability on examination. Clinical assessment included the anterior drawer test and talar tilt test.

All patients underwent preoperative MRI as part of the diagnostic assessment. MRI was reviewed for ATFL integrity, associated CFL pathology, osteochondral lesions, and other concomitant pathology. Concomitant CFL sprain was not considered an exclusion criterion provided that CFL repair or reconstruction was not required. Patients with osteochondral lesions were excluded from the study.

Minor concomitant arthroscopic pathology and procedures that did not alter postoperative weight-bearing or the accelerated rehabilitation protocol were permitted. These included synovitis and minor procedures such as synovectomy and loose-body removal. Patients requiring any concomitant procedure that necessitated modification of postoperative weight-bearing or rehabilitation were excluded.

Inclusion Criteria:

- Age between 20 and 55 years old with chronic lateral ankle instability for more than 3 months
- Failure of conservative management with structured rehabilitation
- Primary arthroscopic ATFL repair

Exclusion Criteria:

- Previous ipsilateral ankle surgery
- Osteochondral lesions
- Concomitant major procedure requiring modification of postoperative weight-bearing or rehabilitation, including cartilage restoration procedures or corrective osteotomy
- Requirement for concomitant CFL repair or reconstruction
- Inflammatory arthropathy

2.2. Surgical technique

All procedures were performed by the senior author of this study. Patients were positioned supine with a thigh tourniquet applied. Standard anteromedial and anterolateral arthroscopic portals were established for diagnostic ankle arthroscopy and assessment of concomitant intra-articular pathology. Associated intra-articular pathology was assessed at diagnostic arthroscopy. Minor concomitant procedures that did not alter postoperative weight-bearing or rehabilitation were permitted.

Lateral ligament repair was performed using the all-arthroscopic modified lasso-loop technique, as previously described by Takao et al. [14]. This technique involves advancement of the remnant anterior talofibular ligament (ATFL), together with the surrounding soft tissues, into the fibular footprint using a suture-relay method. A single 1.8-mm all-suture anchor (Q-Fix; Smith & Nephew) was used in all cases. Final construct stability was assessed arthroscopically to confirm restoration of lateral ankle stability and appropriate ligament tensioning.

2.3. Postoperative rehabilitation protocol

All patients followed a criterion-based standardised accelerated rehabilitation protocol developed at our institution, emphasising early weight-bearing, early range of motion, and progressive functional recovery. Advancement through each phase of rehabilitation was contingent upon achieving predefined functional milestones, with each patient's progress reviewed and assessed individually by the multidisciplinary rehabilitation team.

Immediately following surgery, patients were managed with a wool-and-crepe bandage only. No ankle brace was used, and crutches were not routinely provided. Immediate weight-bearing was permitted from postoperative day 0.

All patients underwent supervised postoperative rehabilitation through the institutional physiotherapy service. Physiotherapy commenced immediately after surgery, with patients initially reviewed approximately once weekly and subsequent frequency adjusted according to clinical progress. Patients were also provided with a structured home exercise programme.

2.3.1. Early Phase (Postoperative Day 0–7)

Patients were allowed immediate weight-bearing and range-of-motion exercises postoperatively, with a focus on dorsiflexion and eversion.

During the first 3 postoperative weeks, patients were instructed not to actively perform plantarflexion or inversion beyond neutral. These movements were excluded from the prescribed exercise programme and reinforced during supervised physiotherapy sessions to protect the repair.

Early strengthening included isometric exercises for the foot and ankle muscles and functional exercises targeting the proximal lower-limb kinetic chain. Gait training was initiated early with emphasis on dorsiflexion during the stance phase.

2.3.2. Intermediate Phase (Postoperative Day 7–28)

Rehabilitation during this phase focused on restoring:

- Dorsiflexion range of motion in weight-bearing
- Dorsiflexion and eversion strength
- Proprioceptive training

Gait symmetry, resistance-band exercises, and proprioceptive training were progressed according to the patient's ability. Concurrently, short foot intrinsic muscle strengthening exercises were introduced to facilitate neuromuscular re-education and restore dynamic foot stability.

At 3 weeks, gentle active plantarflexion and inversion range-of-

motion exercises were introduced under supervision without overpressure. Patients were encouraged to progress within symptom tolerance while avoiding excessive loading.

2.3.3. Strengthening Phase (Weeks 4–6)

Postoperative precautions were gradually eased, allowing progression toward the full range of motion into plantarflexion and inversion. Progressive strengthening was emphasised, focusing on:

- Heel raises
- Closed-chain strengthening
- Lower limb kinetic chain exercises

Balance and proprioception training were progressed further. Cardiovascular conditioning was introduced, including cycling and walking on an inclined treadmill.

2.3.4. Return-to-Activity Phase (≥ 6 Weeks)

Patients progressed toward higher-level functional activities, including:

- Ankle stabilisation and progressive shock-absorption exercises
- Plyometric exercises
- Multiplanar movement training
- Sport-specific drills

Return to jogging was permitted once the following criteria were met:

- ≥ 25 single-leg heel raises through full range
- Ability to jog on the spot for 30 s without symptoms

Progression to sporting activity was subsequently guided by the treating physiotherapist according to symptoms, functional recovery, and achievement of sport-specific rehabilitation milestones.

2.4. Outcome Measures

Primary Outcomes:

- Time to return to jogging
- Foot and Ankle Ability Measure (FAAM) Activities of Daily Living (ADL) and Sports scores preoperatively and at 6 weeks, 3 months, and 6 months.

Secondary Outcomes:

- Visual Analogue Scale (VAS) preoperatively and at 3 and 6 months postoperatively.

Additional functional outcomes:

- Preoperative participation in sport
- Return to pre-injury level of sport
- Time to return to pre-injury level of sport

Safety Outcomes:

- Recurrent instability
- Repair failure
- Wound complications
- Reoperation

2.5. Data Collection

FAAM and VAS scores were routinely recorded at the predefined clinical follow-up timepoints. Data for the present study were subsequently extracted retrospectively from the electronic medical record by

a study investigator and entered into a deidentified study dataset.

Clinical and physiotherapy records were additionally reviewed to determine time to return to jogging, preoperative sporting participation, return to pre-injury sporting level, time to return to pre-injury sporting level, and documented postoperative complications.

All 20 included patients had complete outcome data through the predefined 6-month assessment, with no losses to follow-up for these outcomes.

2.6. Statistical Analysis

Continuous variables were assessed for normality using the Shapiro–Wilk test. As several variables were non-normally distributed and the sample size was small, nonparametric statistical methods were used for repeated outcome measures. Changes in FAAM ADL, FAAM Sports, and VAS scores across timepoints were assessed using the Friedman test. When an overall significant difference was identified, post hoc pairwise comparisons were performed using the Wilcoxon signed-rank test with Holm adjustment for multiple comparisons. Effect sizes for pairwise comparisons were calculated as $r = Z/\sqrt{N}$, and Kendall's coefficient of concordance was reported for Friedman analyses.

Categorical variables, including preoperative sporting participation and return to pre-injury sporting level, were summarised as frequencies and percentages. Time to return to jogging and time to return to pre-injury sporting level were summarised using the median, first and third quartiles (Q1–Q3), and range.

All statistical tests were two-tailed, and statistical significance was set at $p < 0.05$. Analyses were performed using IBM SPSS Statistics, version 27.0.

3. Results

A total of 20 patients were included with a mean follow-up duration of 11.1 ± 3.5 months. Patient demographic characteristics are summarised in Table 1.

All 20 patients underwent preoperative MRI. Concomitant CFL sprain was identified in 12 patients (60.0%); however, none required CFL repair or reconstruction. Patients with osteochondral lesions were excluded from the cohort. No patient underwent microfracture, chondroplasty, or another cartilage resurfacing procedure. Minor concomitant arthroscopic procedures that did not alter postoperative weight-bearing or rehabilitation, such as synovectomy or loose-body removal, were not systematically quantified.

3.1. Functional Outcomes

FAAM Activities of Daily Living scores improved progressively from a median of 76.2 (Q1–Q3, 67.0–83.0) preoperatively to 87.6 (81.2–92.5) at 6 weeks, 94.0 (87.3–98.4) at 3 months, and 99.2 (95.9–100.0) at 6 months. There was a significant difference across all timepoints (Friedman χ^2 [3] = 57.735, $p < 0.001$; Kendall's $W = 0.962$). All post hoc Wilcoxon signed-rank comparisons remained significant following Holm adjustment (all adjusted $p < 0.001$). The preoperative-to-6-month improvement demonstrated a large effect size ($r = 0.88$).

Table 1
Patient demographics and baseline characteristics.

Variable	Value
Age, years (mean $\pm$ SD)	30.1 $\pm$ 10.1
Sex, n (%)	Female: 10 (50.0%) Male: 10 (50.0%)
Laterality, n (%)	Left: 8 (40.0%) Right: 12 (60.0%)
CFL sprain on preoperative MRI, n (%)	12 (60.0%)
Follow-up duration, months (mean $\pm$ SD)	11.1 $\pm$ 3.5

Values are presented as mean $\pm$ standard deviation or number (percentage).

FAAM Sports scores improved progressively from a median of 43.5 (Q1–Q3, 32.2–63.4) preoperatively to 63.5 (45.8–70.0) at 6 weeks, 78.6 (64.6–84.2) at 3 months, and 94.7 (82.8–100.0) at 6 months. There was a significant difference across all timepoints (Friedman χ^2 [3] = 58.303, $p < 0.001$; Kendall's $W = 0.972$). All post hoc Wilcoxon signed-rank comparisons remained significant following Holm adjustment (all adjusted $p < 0.001$), with large effect sizes ranging from $r = 0.83$ – 0.88 . The preoperative-to-6-month improvement demonstrated a large effect size of $r = 0.88$.

The median time to return to jogging was 42 days (Q1–Q3, 35–42 days; range, 21–91 days). All 20 patients returned to jogging.

Seventeen of 20 patients (85.0%) participated in sport preoperatively. All 17 patients who participated in sport preoperatively returned to their pre-injury level of sport during follow-up (17/17, 100%). The median time to return to pre-injury level of sport was 8 weeks (Q1–Q3, 6–11 weeks; range, 6–12 weeks).

3.2. Pain Outcomes

VAS scores decreased from a median of 5.0 (Q1–Q3, 4.0–6.0) preoperatively to 2.0 (1.0–2.0) at 3 months and 0.0 (0.0–1.0) at 6 months. There was a significant difference across all timepoints (Friedman χ^2 [2] = 37.333, $p < 0.001$; Kendall's $W = 0.933$). All post hoc pairwise comparisons remained significant following Holm adjustment (all adjusted $p < 0.001$). The preoperative-to-6-month reduction demonstrated a large effect size ($r = 0.88$). Functional and pain outcomes are presented in Table 2.

3.3. Complications

No documented cases of recurrent instability, repair failure, wound complication, or reoperation were observed during the follow-up period. No other documented postoperative complications were identified on review of the clinical records. Early recovery, return-to-sport, and safety outcomes are summarised in Table 3.

4. Discussion

The principal finding of this study was that accelerated rehabilitation following arthroscopic ATFL repair was associated with significant early improvements in function and pain, with no predefined postoperative complications recorded in this cohort. In addition, all patients who participated in sport preoperatively returned to their pre-injury level of sport, with a median return time of 8 weeks.

Patients demonstrated progressive improvements in FAAM ADL and Sports scores through 6 months. These improvements were accompanied by large within-patient effect sizes across all measured outcomes, indicating substantial changes during the postoperative period.

A key strength of this study is the demonstration of early functional recovery, with patients returning to jogging at a median of approximately 6 weeks postoperatively. Jogging was used as an early, reproducible functional milestone and should not be interpreted as unrestricted return to sport or return to preinjury sporting level. Importantly, return to sport was assessed separately in the present cohort. Seventeen patients participated in sport preoperatively, and all

Table 3

Early Functional Recovery, Return-to-Sport, and Safety Outcomes.

Outcome	Value
Time to jogging, days, median (Q1–Q3)	42 (35–42)
Time to jogging, range, days	21–91
Returned to jogging, n (%)	20 (100%)
Participated in sport preoperatively, n (%)	17 (85.0%)
Returned to pre-injury level of sport, n/N (%)	17/17 (100%)
Return to pre-injury level of sport, weeks, median (Q1–Q3)	8 (6–11)
Time to return to pre-injury level of sport, range, weeks	6–12
Any recorded complication, n (%)	0 (0%)
Recurrent instability, n (%)	0 (0%)
Repair failure, n (%)	0 (0%)
Wound complication, n (%)	0 (0%)
Reoperation, n (%)	0 (0%)

17 returned to their pre-injury level of sport at a median of 8 weeks (Q1–Q3, 6–11 weeks; range, 6–12 weeks). This distinction between return to jogging and return to pre-injury sporting level provides a more clinically meaningful assessment of functional recovery. These findings are consistent with previous reports supporting early functional rehabilitation following lateral ankle ligament repair. Teramoto et al. reported early return to sport following accelerated rehabilitation after arthroscopic lateral ankle ligament repair [13]. The present study adds to this evidence by describing serial FAAM ADL and Sports outcomes, pain scores, return to jogging, and return to pre-injury sporting level following a standardised, criterion-based rehabilitation protocol after isolated arthroscopic ATFL repair.

The present rehabilitation protocol differed from more protective postoperative strategies by permitting immediate weight-bearing from postoperative day 0 without an ankle brace or routine crutch use, while restricting potentially provocative plantarflexion and inversion during the initial healing period. Progression thereafter was criterion-based and supervised by physiotherapy. The favourable early functional outcomes observed in this cohort suggest that such a structured approach may be feasible after isolated arthroscopic ATFL repair in appropriately selected patients.

Martin et al. reported favourable early functional outcomes with no recurrent instability or revision surgery following immediate weight-bearing as part of an accelerated rehabilitation protocol after suture-tape-augmented modified Broström repair [12]. In contrast, the present cohort underwent isolated arthroscopic ATFL repair without suture tape augmentation, suggesting that favourable early recovery may also be achievable without additional augmentation. However, the absence of a comparative group prevents conclusions regarding equivalence or superiority between augmented and non-augmented constructs.

Despite concerns that early weight-bearing and mobilisation may predispose to repair elongation or failure, no cases of recurrent instability, repair failure, or other documented postoperative complications were observed during the available follow-up. These findings are consistent with the low complication rates reported in previous studies of early rehabilitation after lateral ankle ligament repair [12,13]. Collectively, these observations suggest that a controlled accelerated rehabilitation protocol may be feasible in appropriately selected patients, although comparative studies are required to determine its safety relative to conventional rehabilitation.

Table 2

Functional and pain outcomes following accelerated rehabilitation after ATFL repair.

Outcome	Preoperative	6 Weeks	3 Months	6 Months	Overall p-value
FAAM ADL, median (Q1–Q3)	76.2 (67.0–83.0)	87.6 (81.2–92.5)	94.0 (87.3–98.4)	99.2 (95.9–100.0)	< 0.001
FAAM Sports, median (Q1–Q3)	43.5 (32.2–63.4)	63.5 (45.8–70.0)	78.6 (64.6–84.2)	94.7 (82.8–100.0)	< 0.001
VAS, median (Q1–Q3)	5.0 (4.0–6.0)	–	2.0 (1.0–2.0)	0.0 (0.0–1.0)	< 0.001

Values are presented as median (first–third quartile). Overall differences across timepoints were assessed using the Friedman test. Post hoc pairwise comparisons were performed using the Wilcoxon signed-rank test with Holm adjustment for multiple comparisons. FAAM = Foot and Ankle Ability Measure; ADL = Activities of Daily Living; VAS = Visual Analogue Scale.

Concomitant pathology was also considered in the present cohort. All patients underwent preoperative MRI, and 12 of 20 patients (60.0%) demonstrated concomitant CFL sprain; however, none required CFL repair or reconstruction. Patients with osteochondral lesions were excluded. Minor concomitant arthroscopic pathology and procedures that did not alter postoperative weight-bearing or rehabilitation, including synovitis, synovectomy, and loose-body removal, were permitted. Given the small cohort and the fact that these minor procedures were not systematically quantified, their independent influence on postoperative outcomes could not be assessed.

Functional outcomes continued to improve through 6 months, suggesting that early mobilisation did not prevent continued short-term recovery. However, longer-term follow-up is required to determine the durability of these improvements and the incidence of late recurrent instability.

4.1. Limitations

This study has several limitations. First, its retrospective design introduces the potential for selection bias. Second, the absence of a control group limits direct comparison with conventional rehabilitation protocols. Third, the relatively small sample size restricts the generalisability of the findings. Additionally, the short-term follow-up duration was sufficient only to assess early outcomes; longer-term follow-up is required to evaluate the durability of results and late complications.

Return-to-sport outcomes were obtained retrospectively from clinical and physiotherapy documentation rather than using a prospectively administered validated return-to-sport instrument, which may introduce documentation bias. Furthermore, the relatively small cohort precluded meaningful subgroup analysis according to concomitant CFL pathology or minor arthroscopic procedures. Minor concomitant procedures were not systematically quantified because they did not alter postoperative weight-bearing or rehabilitation. Finally, patients with osteochondral lesions were excluded, and therefore the findings may not be generalisable to patients requiring concomitant cartilage procedures or other interventions that necessitate modification of postoperative rehabilitation.

5. Conclusion

Accelerated rehabilitation following isolated arthroscopic ATFL repair was associated with significant early improvements in patient-reported outcomes, a median return to jogging of 42 days and return to pre-injury sporting level in all preoperatively active patients. No documented repair failures or recurrent instability were observed during short-term follow-up. These findings suggest the short-term feasibility of a criterion-based accelerated rehabilitation protocol following isolated arthroscopic ATFL repair. Further prospective comparative studies with longer follow-up are warranted to assess durability and recurrent instability, and longer-term return-to-sport outcomes.

CRediT authorship contribution statement

Dhivakaran Gengatharan: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Zachary Poon Qi Jing:** Writing – review & editing, Writing – original draft, Visualization, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hon Keng Pui Bryan:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Melissa Chan Wen Yi:** Writing – review & editing, Visualization,

Validation, Data curation, Conceptualization. **Elizabeth Goh Tien Nee:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Data curation, Conceptualization. **Lim Joey:** Writing – review & editing, Software, Resources, Formal analysis, Data curation, Conceptualization. **Eric Cher Wei Liang:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Ethics Statement

Ethical approval for this study was waived by the SingHealth Centralised Institutional Review Board.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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